Lab Environment Overview — Simulated Enterprise IT Support Lab

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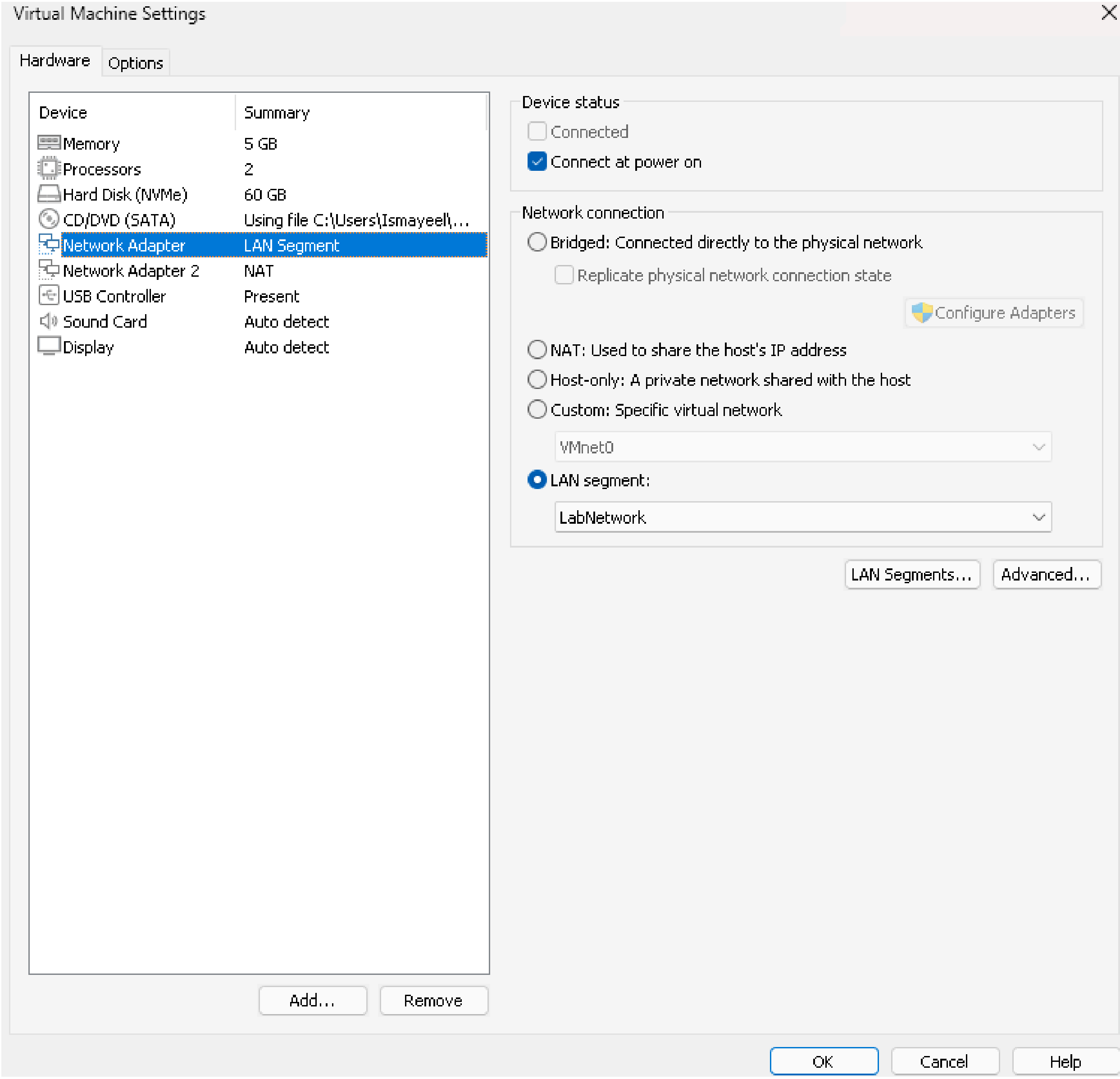
Section 1: Project Introduction

This document provides a technical overview of the simulated enterprise IT support lab environment built as part of Project 2. The objective of this lab was to replicate a real-world business IT infrastructure using virtualization technology, allowing for the practical simulation of helpdesk support scenarios, troubleshooting workflows, and IT service desk operations.

Section 2: Virtualization Platform

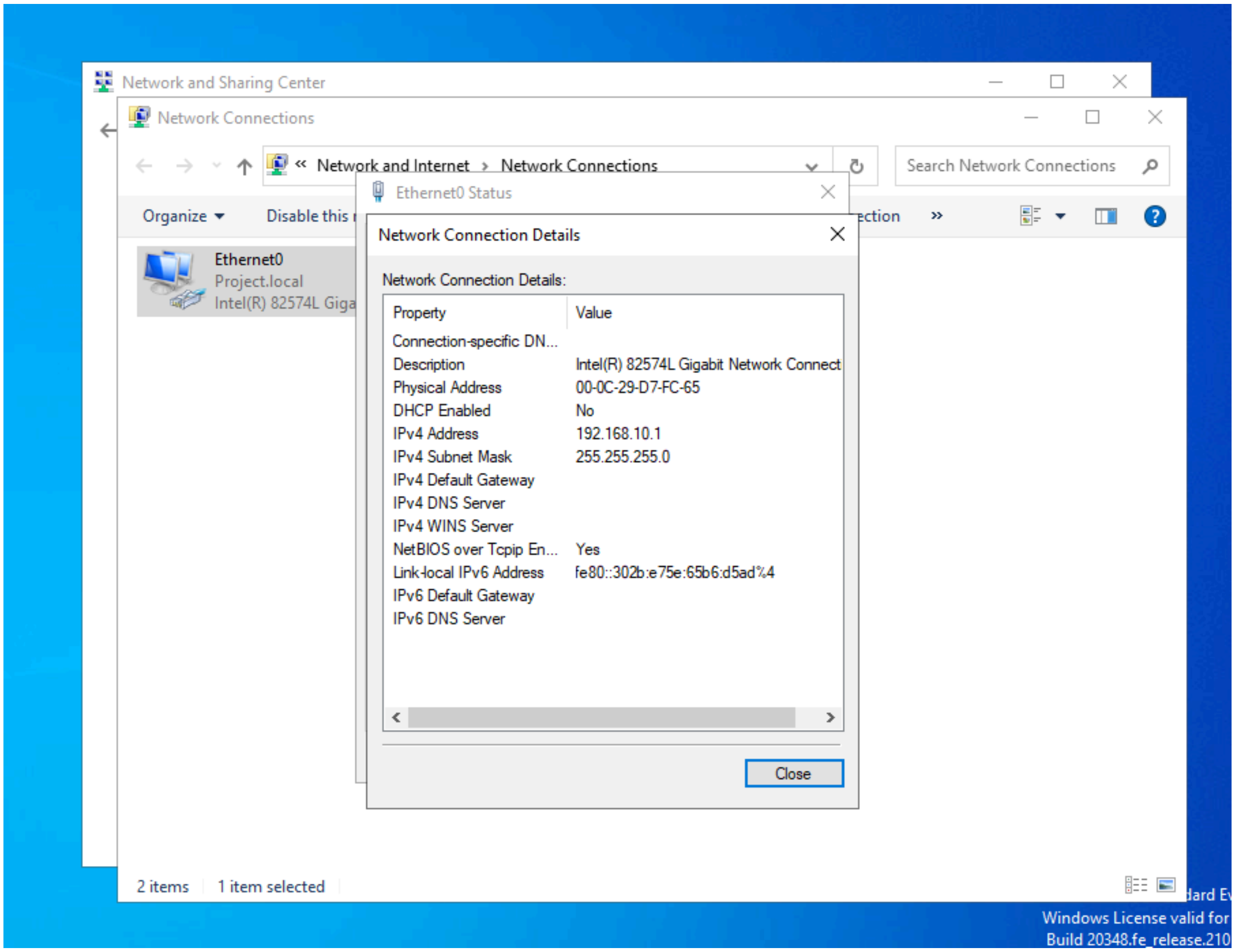
The lab was built using VMware Workstation 17 Player running on a Windows host machine with 16GB of RAM. Two virtual machines were deployed to simulate a small business network environment consisting of a server and a client workstation.

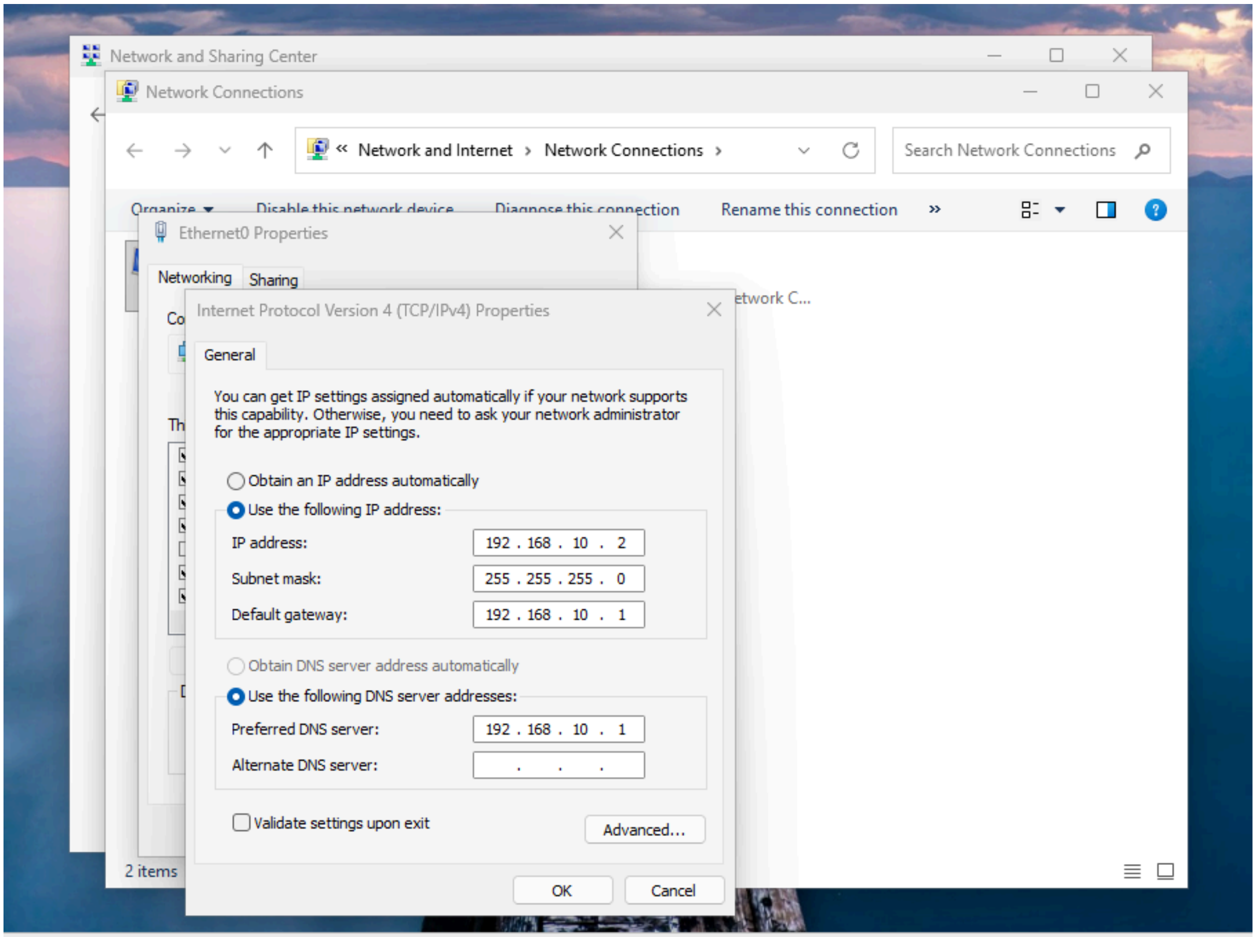


Section 3: Virtual Machine Configuration

The first virtual machine runs Windows Server 2022 and serves as the domain controller for the lab environment. It hosts Active Directory Domain Services under the domain Project.local, and also runs the Spiceworks Cloud Help Desk ticketing system used throughout the lab. The server was configured with a static IP address of 192.168.10.1 on the internal Host-Only network adapter.

The second virtual machine runs Windows 11 Pro and represents a standard employee workstation. It was successfully joined to the Project.local domain and configured with a static IP address of 192.168.10.2. This machine was used to simulate end user issues and remote support scenarios throughout the lab.

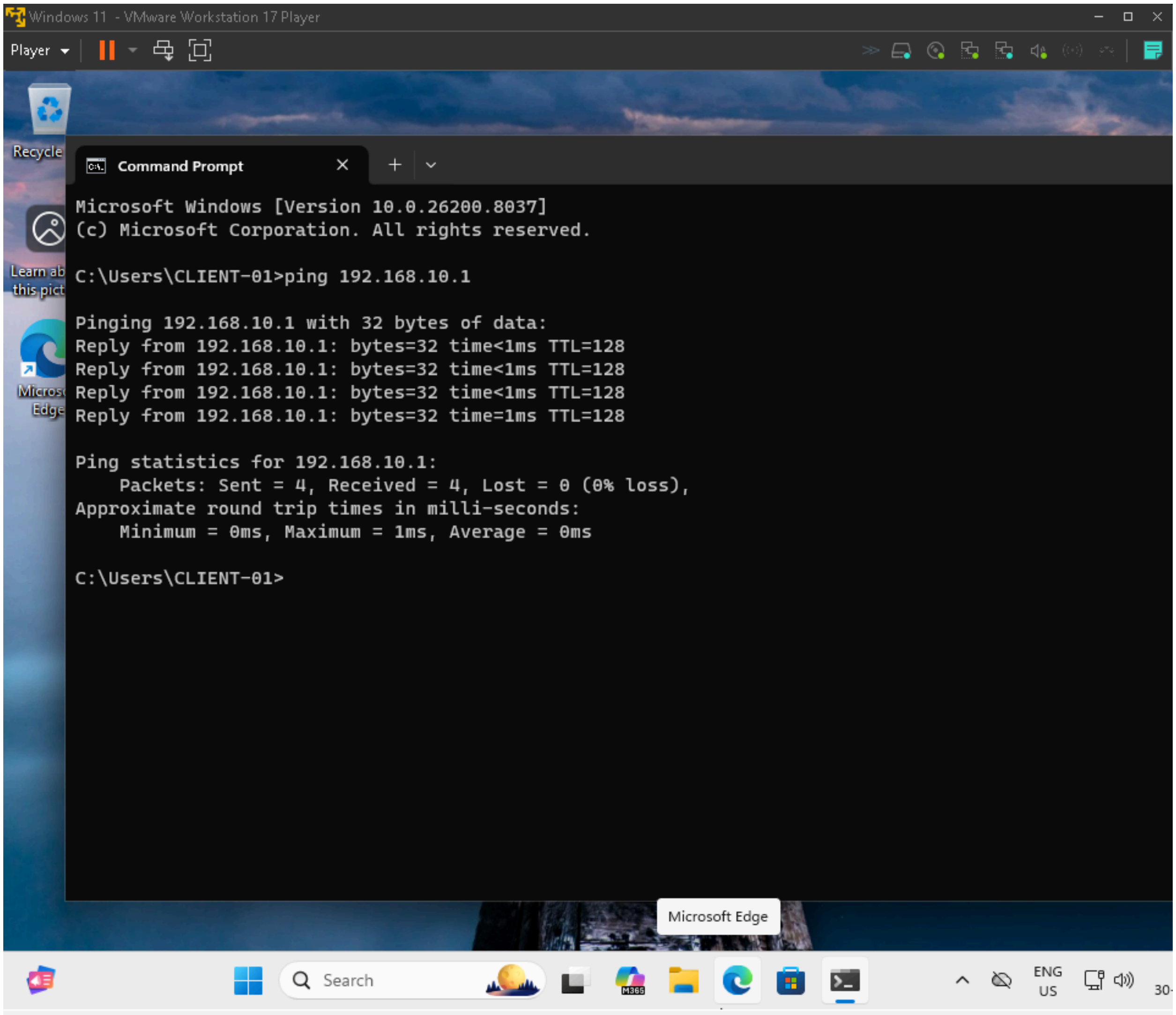

Network and Sharing Center
Network Connections
Network and Internet > Network Connections
Search Network Connections
Organize
Disable this
Ethernet0 Status
Network Connection Details
Ethernet0
Project.local
Intel(R) 82574L Giga
Network Connection Details:
Property
Value
Connection-specific DN...
Description
Intel(R) 82574L Gigabit Network Connect
Physical Address
00-0C-29-D7-FC-65
DHCP Enabled
No
IPv4 Address
192.168.10.1
IPv4 Subnet Mask
255.255.255.0
IPv4 Default Gateway
IPv4 DNS Server
IPv4 WINS Server
NetBIOS over Tcpip En...
Yes
Link-local IPv6 Address
fe80::302b:e75e:65b6:d5ad%4
IPv6 Default Gateway
IPv6 DNS Server
Close
2 items
1 item selected
Windows License valid for
Build 20348.fe_release.210

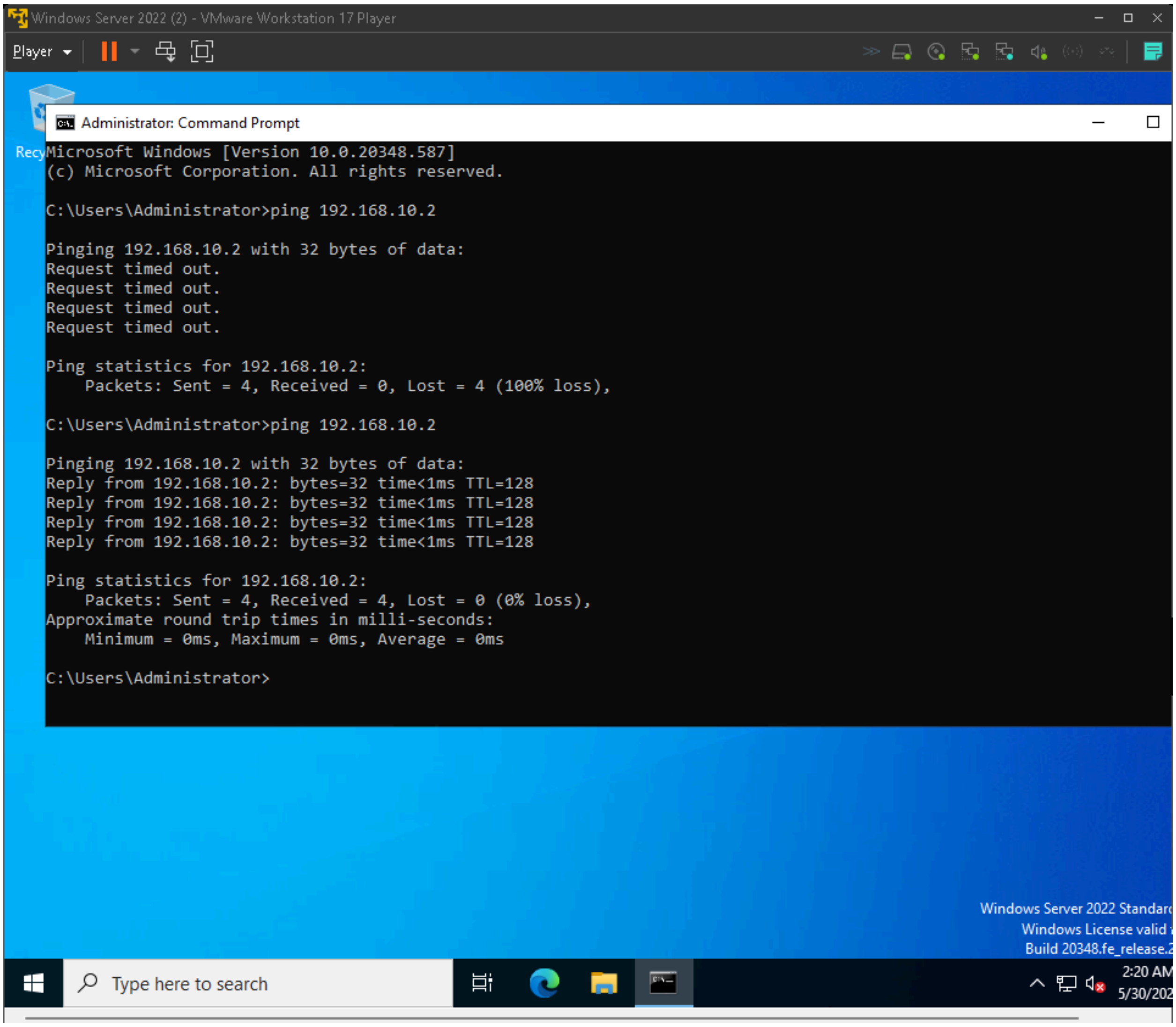


Section 4: Network Configuration

Both virtual machines were configured with two network adapters each. The first adapter uses NAT to provide internet access to both machines. The second adapter uses a custom LAN Segment named LabNetwork to create a private internal network between the two virtual machines, simulating a local business network environment.

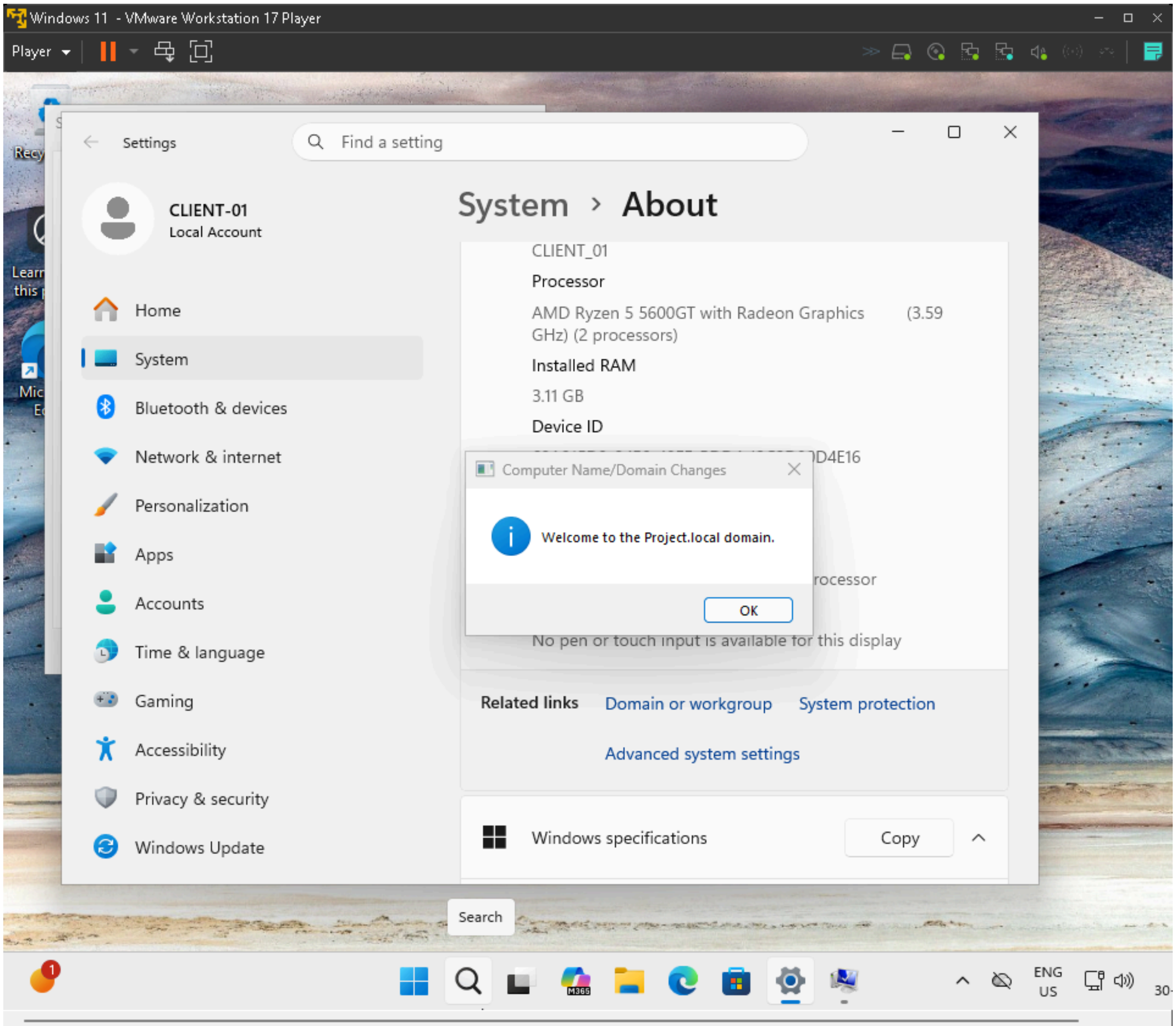
Network connectivity between both machines was verified using ping tests in both directions. The Windows Server successfully pinged the Windows 11 client at 192.168.10.2, and the Windows 11 client successfully pinged the server at 192.168.10.1, both with 0% packet loss.

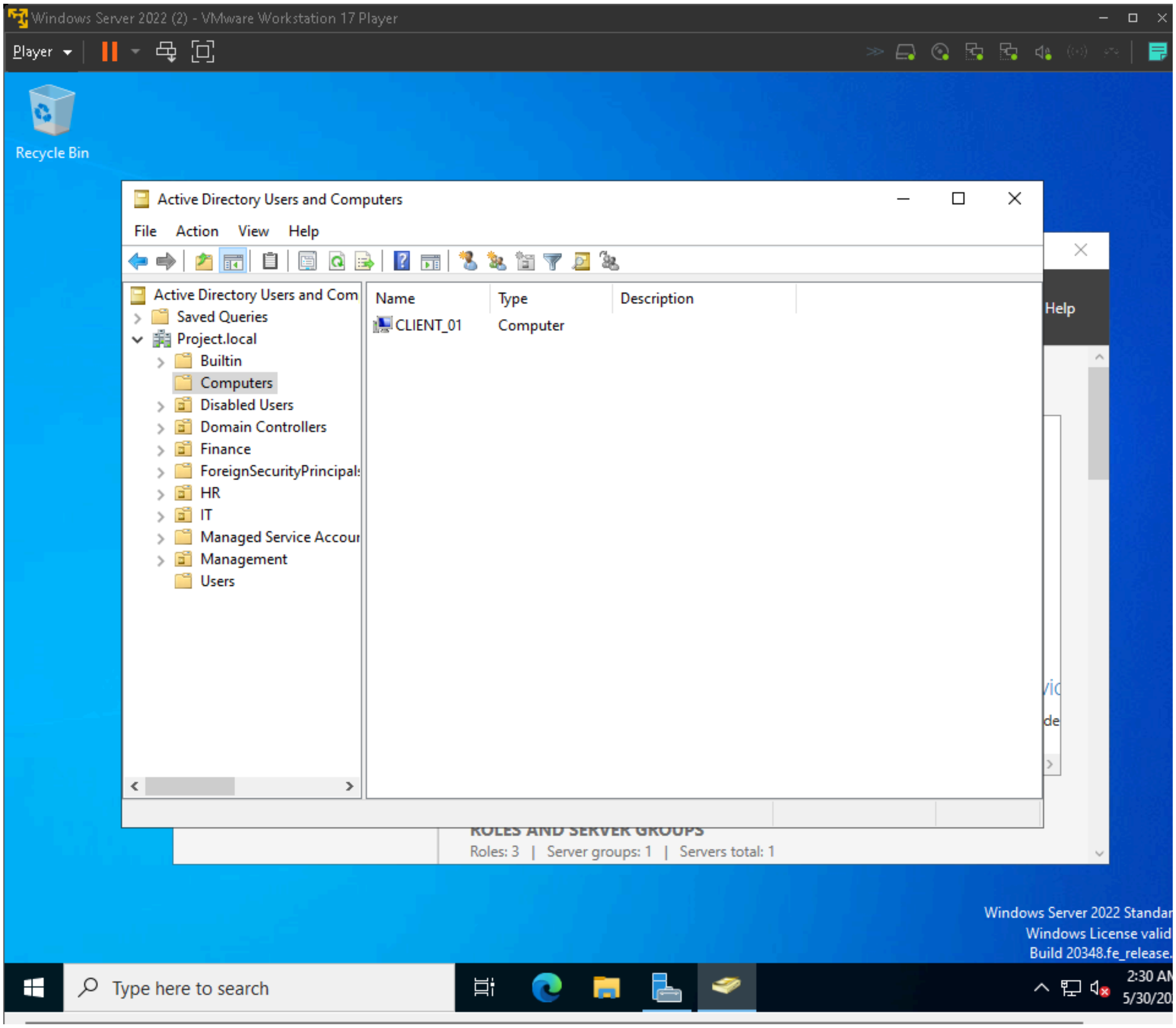

Windows 11 - VMware Workstation 17 Player
Player
Recycle
Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.
C:\Users\CLIENT-01>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Users\CLIENT-01>
Microsoft Edge
Search
ENG
US



Section 5: Active Directory Domain

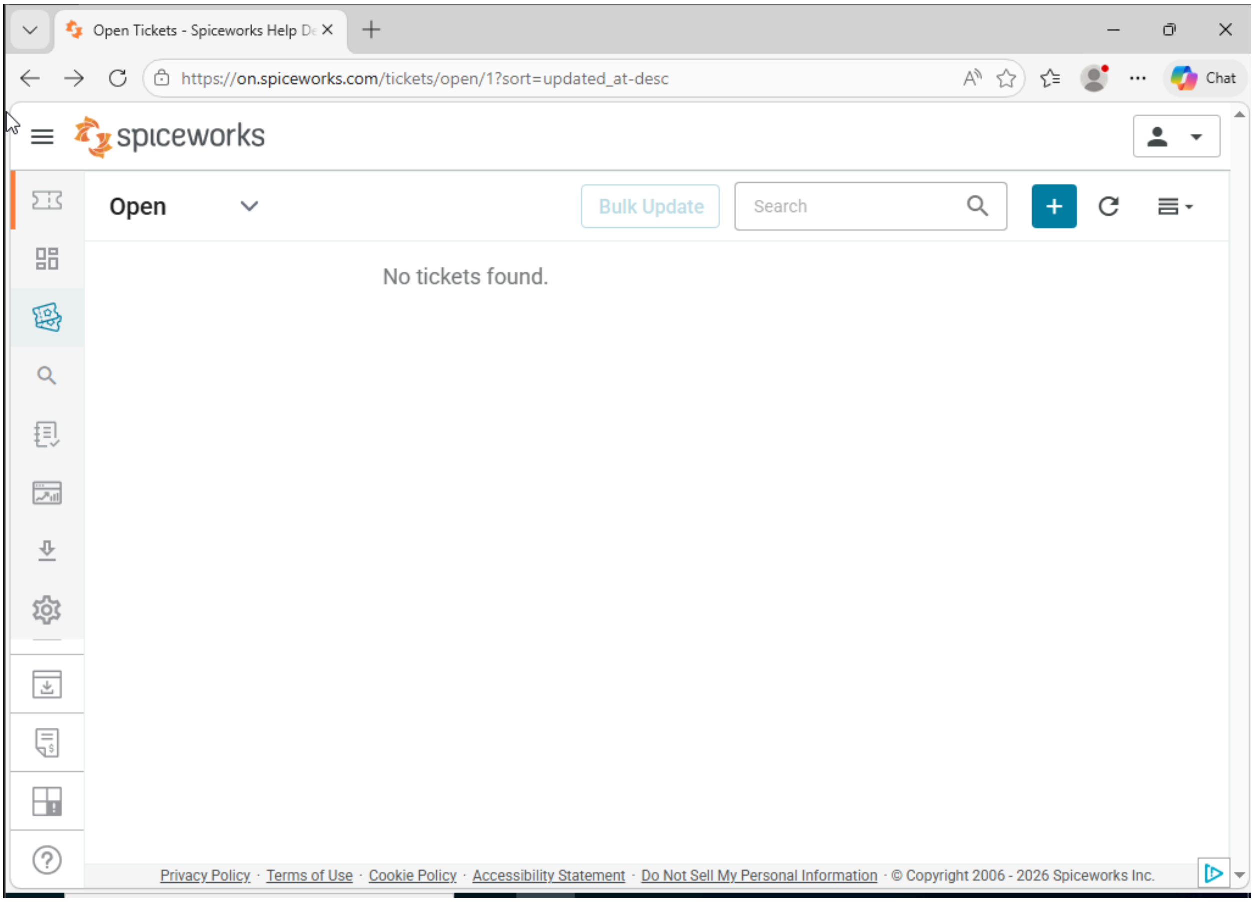
The Windows Server 2022 virtual machine was configured as a domain controller for the Project.local Active Directory domain. This domain was originally built during Project 1 and includes a full organisational unit hierarchy, security groups, user accounts, and group policy objects. The Windows 11 client machine was successfully joined to this domain, confirmed by the appearance of CLIENT_01 in the Active Directory Computers container on the server.


Windows 11 - VMware Workstation 17 Player
Player
Settings
Find a setting
CLIENT-01
Local Account
Home
System
Bluetooth & devices
Network & internet
Personalization
Apps
Accounts
Time & language
Gaming
Accessibility
Privacy & security
Windows Update
System › About
CLIENT_01
Processor
AMD Ryzen 5 5600GT with Radeon Graphics (3.59 GHz) (2 processors)
Installed RAM
3.11 GB
Device ID
Computer Name/Domain Changes
Welcome to the Project.local domain.
OK
No pen or touch input is available for this display
Related links
Domain or workgroup
System protection
Advanced system settings
Windows specifications
Copy
Search
ENG
US



Section 6: Helpdesk Ticketing System

Spiceworks Cloud Help Desk was deployed as the ticketing platform for this lab. The system was accessed via browser on the Windows Server virtual machine and configured with four ticket rules for automatic categorisation covering Network, Software, Hardware, and Account issues. Three canned responses were also created to simulate professional helpdesk communication including an Initial Response, Issue Resolved, and Escalation Notice template.


Open Tickets - Spiceworks Help De
https://on.spiceworks.com/tickets/open/1?sort=updated_at-desc
Chat
spiceworks
Open
Bulk Update
Search
No tickets found.
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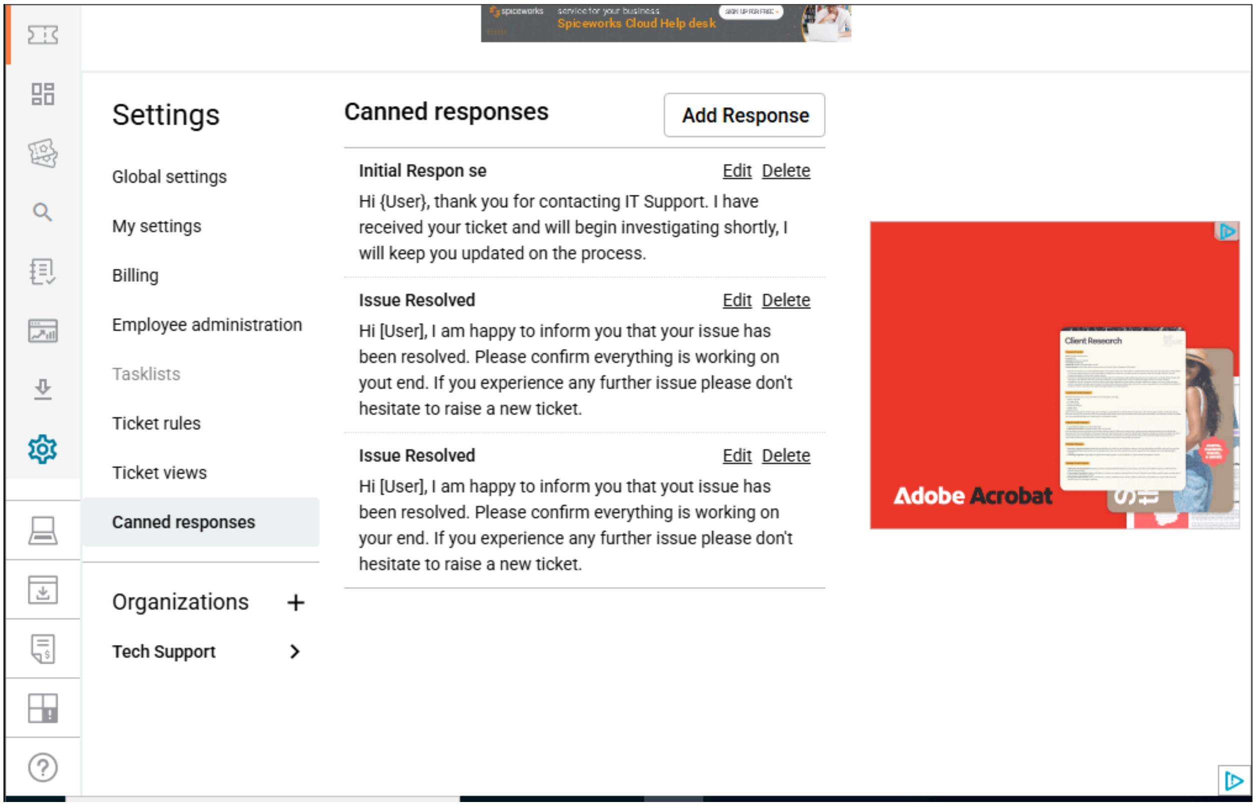
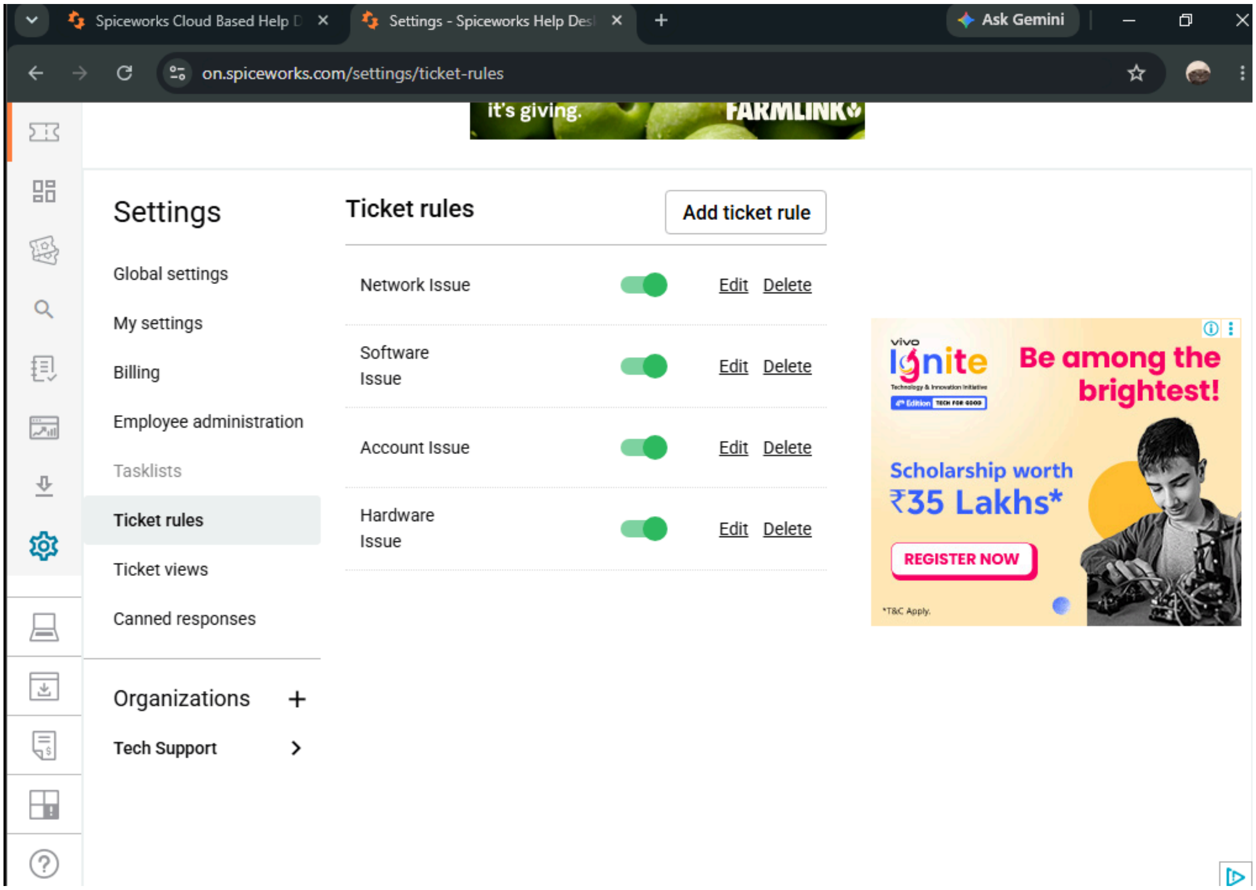

Spiceworks Cloud Help desk
Settings
Global settings
My settings
Billing
Employee administration
Tasklists
Ticket rules
Ticket views
Canned responses
Organizations +
Tech Support
Canned responses
Add Response
Initial Respon se
Edit Delete
Hi {User}, thank you for contacting IT Support. I have received your ticket and will begin investigating shortly, I will keep you updated on the process.
Issue Resolved
Edit Delete
Hi [User], I am happy to inform you that your issue has been resolved. Please confirm everything is working on yout end. If you experience any further issue please don't hesitate to raise a new ticket.
Issue Resolved
Edit Delete
Hi [User], I am happy to inform you that yout issue has been resolved. Please confirm everything is working on your end. If you experience any further issue please don't hesitate to raise a new ticket.
Client Research
Adobe Acrobat



Spiceworks Cloud Based Help D
Settings - Spiceworks Help Des
Ask Gemini
on.spiceworks.com/settings/ticket-rules
Settings
Global settings
My settings
Billing
Employee administration
Tasklists
Ticket rules
Ticket views
Canned responses
Organizations
Tech Support
Ticket rules
Add ticket rule
Network Issue
Edit
Delete
Software Issue
Edit
Delete
Account Issue
Edit
Delete
Hardware Issue
Edit
Delete
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